The one-column spectral minimax identity

A special case of TR-03, not a full resolution

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Status. The argument below proves the $k = 1$ case for every positive spectrum. It does not determine the joint dependence on (n, k) requested in TR-03. This elementary identity may already be implicit in earlier work; no priority or novelty claim is made.

1 Statement

Let $n \geq 3$ and $\lambda = (\lambda_1, \dots, \lambda_n) \in (0, \infty)^n$. Put $\Lambda = \operatorname{diag}(\lambda)$ and $K_V = V^T \Lambda V$ for $V \in O(n)$. Define

$$x_1(\lambda) = \max_{V \in O(n)} \min_{1 \leq i \leq n} \operatorname{tr}\left(K_V - \frac{(K_V)_{:i}(K_V)_{i:}}{(K_V)_{ii}} \right), \qquad y_1(\lambda) = \frac{2e_2(\lambda)}{e_1(\lambda)}.$$

These are the $k = 1$ quantities in the repository target [1].

Theorem 1. *For every positive spectrum,*

$$x_1(\lambda) = y_1(\lambda) = \sum_{j=1}^{n} \lambda_j - \frac{\sum_{j=1}^{n} \lambda_j^2}{\sum_{j=1}^{n} \lambda_j}.$$

Consequently $R_{n,1} = \sup_{\lambda > 0} y_1(\lambda)/x_1(\lambda) = 1$.

2 A zero-diagonal orthogonal basis

Lemma 2. *Every real symmetric matrix H of trace zero has an orthonormal basis $v_1, \dots, v_n$ satisfying $v_i^T H v_i = 0$ for every i.*

Proof. Induct on the dimension. The one-dimensional case is immediate. If $H = 0$, every basis works. Otherwise trace zero implies the presence of a positive and a negative eigenvalue. A unit vector v_1 in the span of corresponding eigenvectors has zero Rayleigh quotient by continuity (or by choosing the squared component weights inversely proportional to the absolute eigenvalues). The compression of H to $v_1^\perp$ is symmetric and has trace $\operatorname{tr} H - v_1^T H v_1 = 0$. Apply induction to that compression and append v_1. □

3 Proof of the identity

Write $S_1 = \sum_j \lambda_j$, $S_2 = \sum_j \lambda_j^2$ and $c = S_2/S_1$. For any orthogonal V and $K = K_V$, the error of choosing column i is

$$E_i(K) = S_1 - \frac{(K^2)_{ii}}{K_{ii}}.$$

The weights K_{ii}/S_1 are positive and sum to one. Hence

$$\min_i E_i(K) \le \sum_i \frac{K_{ii}}{S_1} E_i(K) = S_1 - \frac{S_2}{S_1} = y_1(\lambda).$$

This proves $x_1 \le y_1$.

For the reverse inequality, set $H = \Lambda^2 - c\Lambda$. Its trace is zero. By the lemma, choose an orthogonal matrix V whose columns have zero H-Rayleigh quotient. Since $(V^T \Lambda V)^2 = V^T \Lambda^2 V$, the resulting K obeys

$$(K^2)_{ii} = cK_{ii} \qquad (1 \le i \le n).$$

Thus *every* column has the same error $E_i(K) = S_1 - c = y_1(\lambda)$. Therefore $x_1 \ge y_1$, proving equality. The formula for y_1 follows from $2e_2 = S_1^2 - S_2$. Positivity of the spectrum and $n \ge 3$ ensure $y_1 > 0$, so the ratio is well defined.

4 What remains

For $2 \le k \le n-2$, neither this argument nor the accompanying numerical examples determine the sharp spectral minimax ratio. Simultaneous equalization of all k-column residuals is substantially different from equalizing the diagonal of a single trace-zero symmetric matrix. This note must not be submitted or counted as a resolution of the full TR-03 target.

The script `references/holden-tr03-2026-09-12/verify_TR03.py` checks one radical-valued example exactly and constructs zero-diagonal bases for a fixed set of numerical spectra. These computations are illustrative; the proof above is independent of them.

References

[1] OpenProblemsInNLA, *TR-03: Sharp gap between volume sampling and the worst matrix with a prescribed spectrum*, accessed 12 September 2026.
`https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/randomized-and-low-rank-approximation/TR-03`